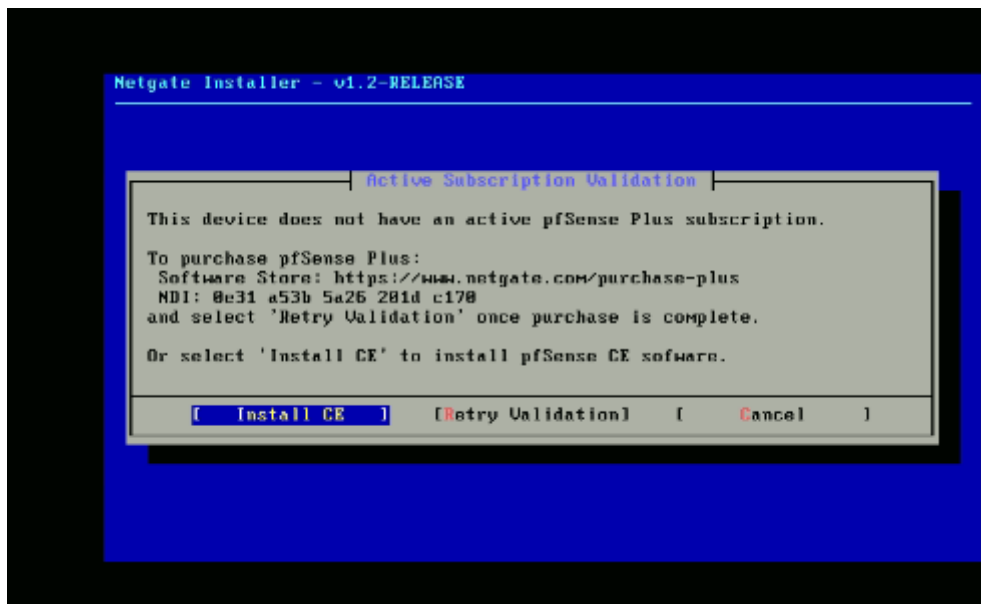


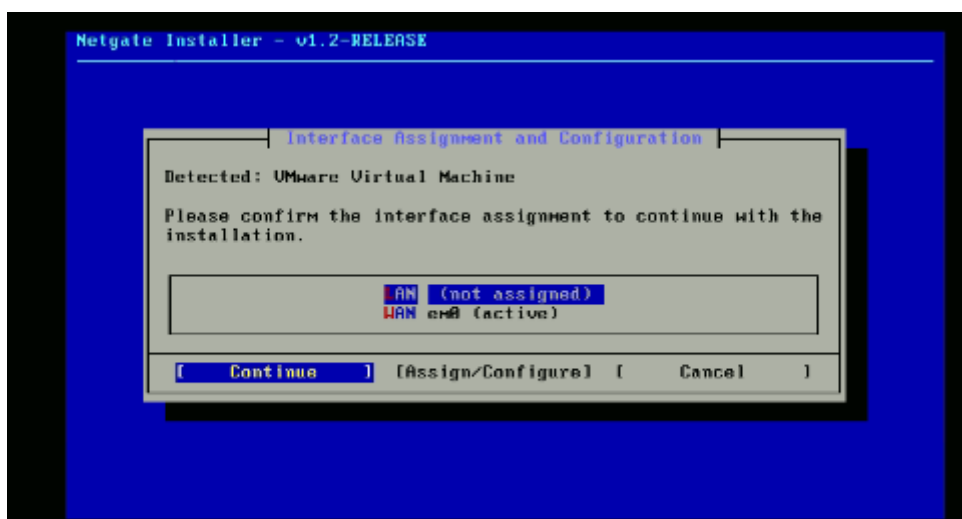
1. Sélection de la version d'installation

- **Description :** Lors de cette étape, nous avons sélectionné la version pfSense CE (Community Edition) pour procéder à l'installation.



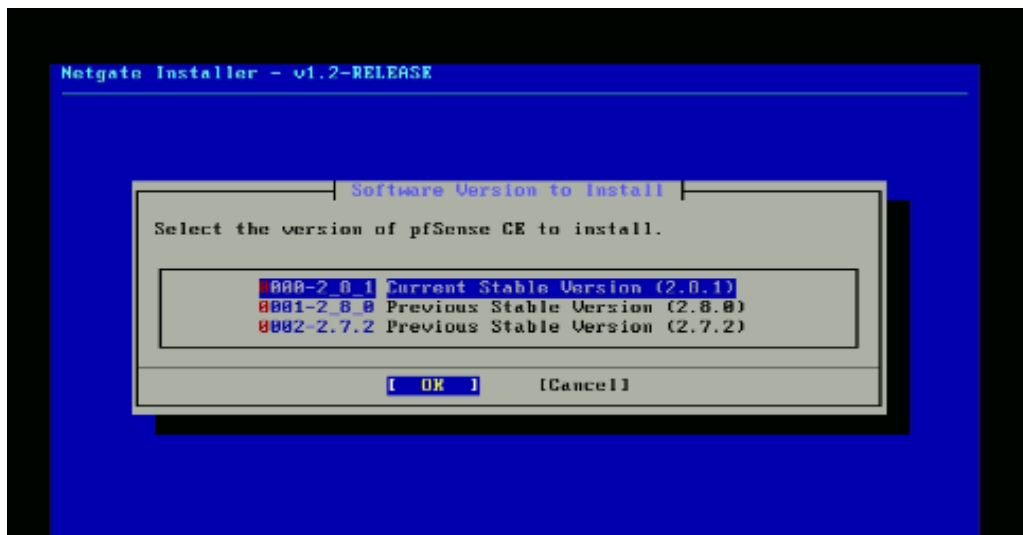
2. Confirmation de l'assignation des interfaces

- **Description :** L'interface WAN (em0) a été détectée comme active, et nous avons confirmé l'assignation pour poursuivre l'installation.



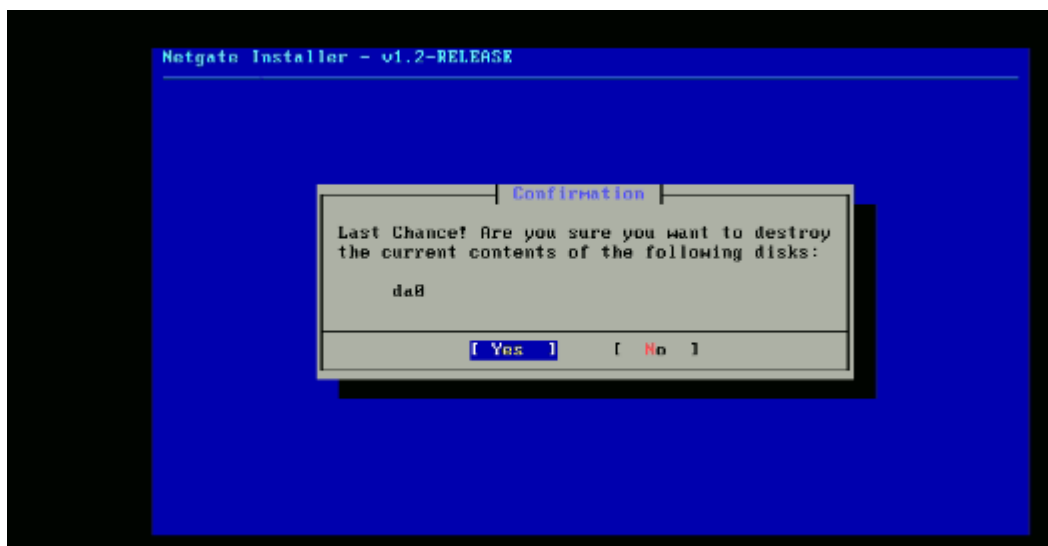
3. Validation de l'abonnement

- **Description :** Comme l'appareil ne dispose pas d'un abonnement pfSense Plus, nous avons choisi l'option "Install CE" pour installer la version communautaire.



4. Initialisation du disque dur

- **Description :** Nous avons confirmé l'effacement des données sur le disque (da0) afin de démarrer une installation propre.



5. Console système après installation

- **Description :** Une fois l'installation terminée, la console principale de pfSense est devenue opérationnelle pour la gestion en ligne de commande.

```

Generating RRD graphs...done.
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 0e31a53b5a26281dc178

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.18.98/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: SSSSS

```

6. Configuration de l'interface réseau local (LAN)

- **Description :** Nous avons sélectionné l'option (2) pour modifier les paramètres de l'interface LAN et avons assigné une adresse IP statique (192.168.100.1).

```

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.18.98/24
LAN (lan) -> em1 ->

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n)

```

7. Finalisation de la configuration

- **Description :** La nouvelle adresse IP a été appliquée avec succès, permettant l'accès à l'interface d'administration via un navigateur web.

```
Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.100.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

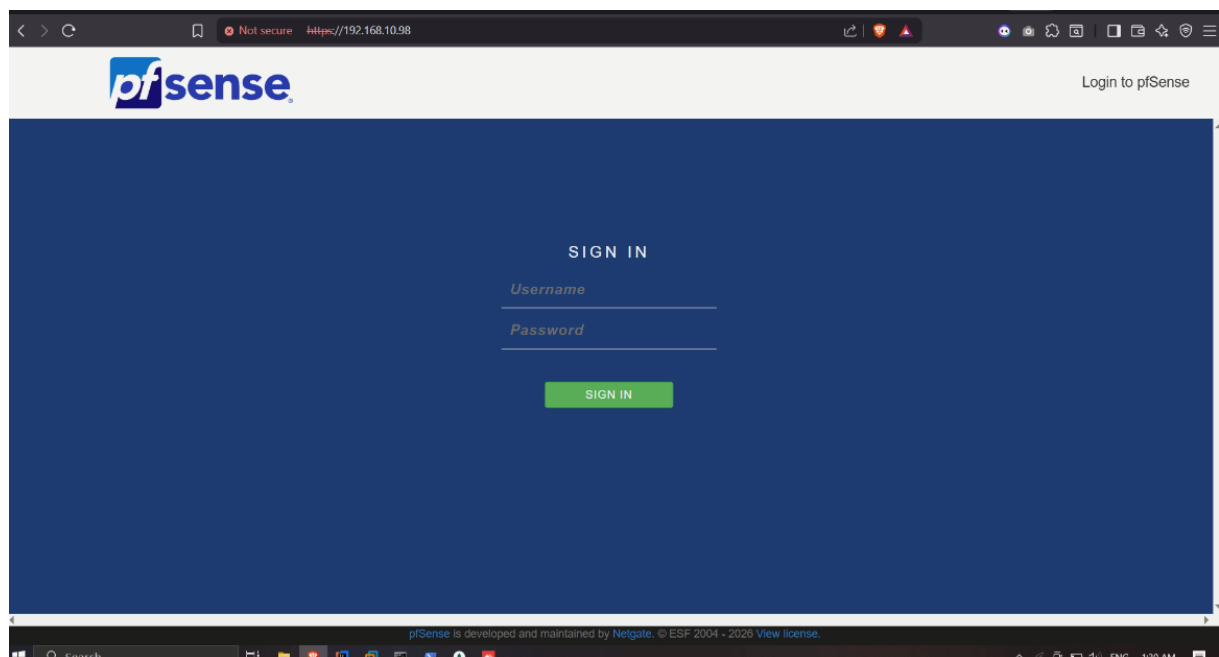
Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

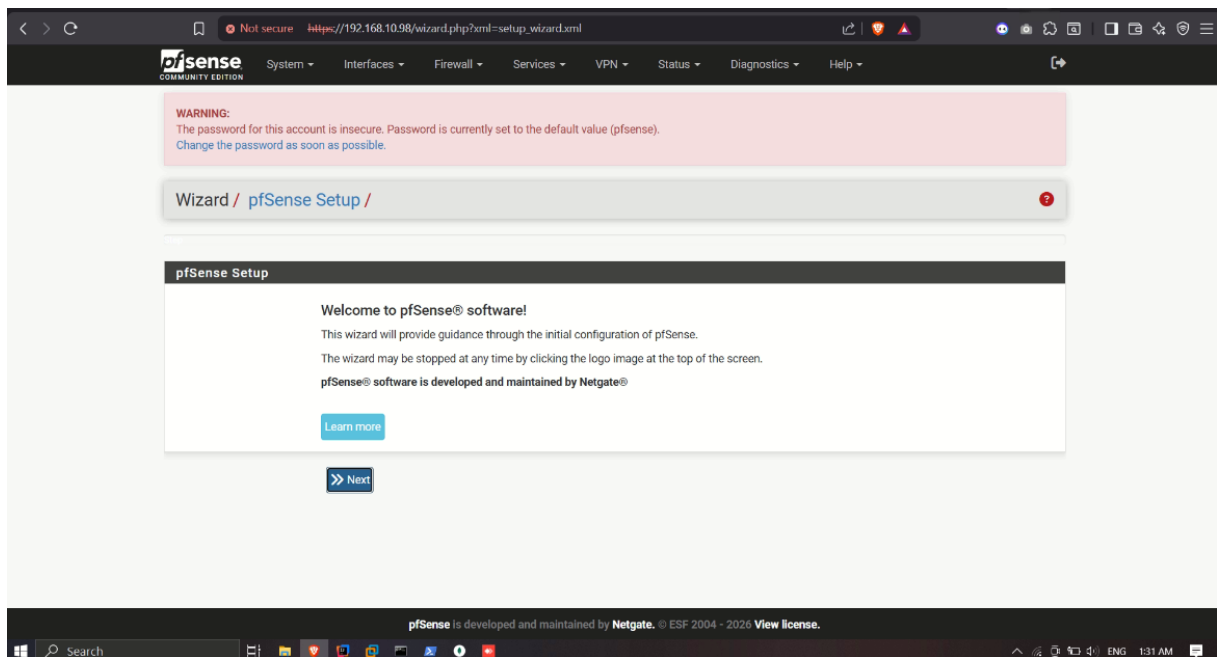
Do you want to revert to HTTP as the webConfigurator protocol? (y/n) █
```

8. Accès à l'interface d'administration (WebConfigurator)

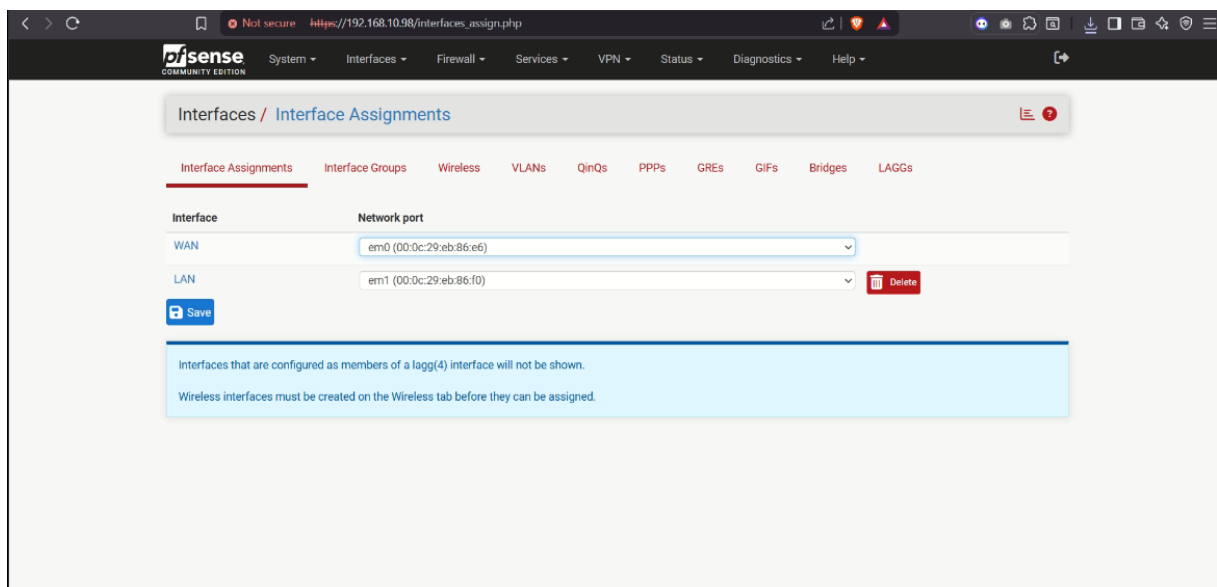
- **Description :** L'image finale montre l'interface de connexion à pfSense via le navigateur, utilisant l'adresse IP configurée.



9. Lancement de l'assistant de configuration (Setup Wizard) : Démarrage de l'assistant initial pour guider la configuration de base de pfSense.



10. Assignment des interfaces : Vérification et confirmation de l'assignation des ports physiques (WAN sur em0 et LAN sur em1).



11. Configuration générale de l'interface LAN : Paramétrage de l'interface LAN (em1) avec une adresse IP statique 192.168.100.1 et un masque de sous-réseau /24.

pfSense COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Interfaces / LAN (em1)

General Configuration

Enable ☒ Enable interface

Description
Enter a description (name) for the interface here.

IPv4 Configuration Type

IPv6 Configuration Type

MAC Address
This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address /

IPv4 Upstream gateway [+ Add a new gateway](#)
If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a WAN type interface.
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

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12. Configuration générale de l'interface WAN : Paramétrage de l'interface WAN (em0) en mode DHCP pour la connexion internet.

Interfaces / WAN (em0)

General Configuration

Enable ☒ Enable interface

Description WAN
Enter a description (name) for the interface here.

IPv4 Configuration Type DHCP

IPv6 Configuration Type DHCP6

MAC Address xxxxxxxxxx
This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xxxxxxxxxx or leave blank.

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex Default (no preference, typically autoselect)
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

DHCP Client Configuration

Options ☐ Advanced Configuration ☐ Configuration Override
Use advanced DHCPv6 configuration options. Override the configuration from this file.

Use IPv4 connectivity as parent interface ☐ Request a IPv6 prefix/information through the IPv4 connectivity link

Request only an IPv6 prefix ☐ Only request an IPv6 prefix, do not request an IPv6 address

DHCPv6 Prefix Delegation size 64
The value in this field is the delegated prefix length provided by the DHCPv6 server. Normally specified by the ISP.

Send IPv6 prefix hint ☐ Send an IPv6 prefix hint to indicate the desired prefix size for delegation

Do not wait for a RA ☐ Required by some ISPs, especially those not using PPPoE

Reserved Networks

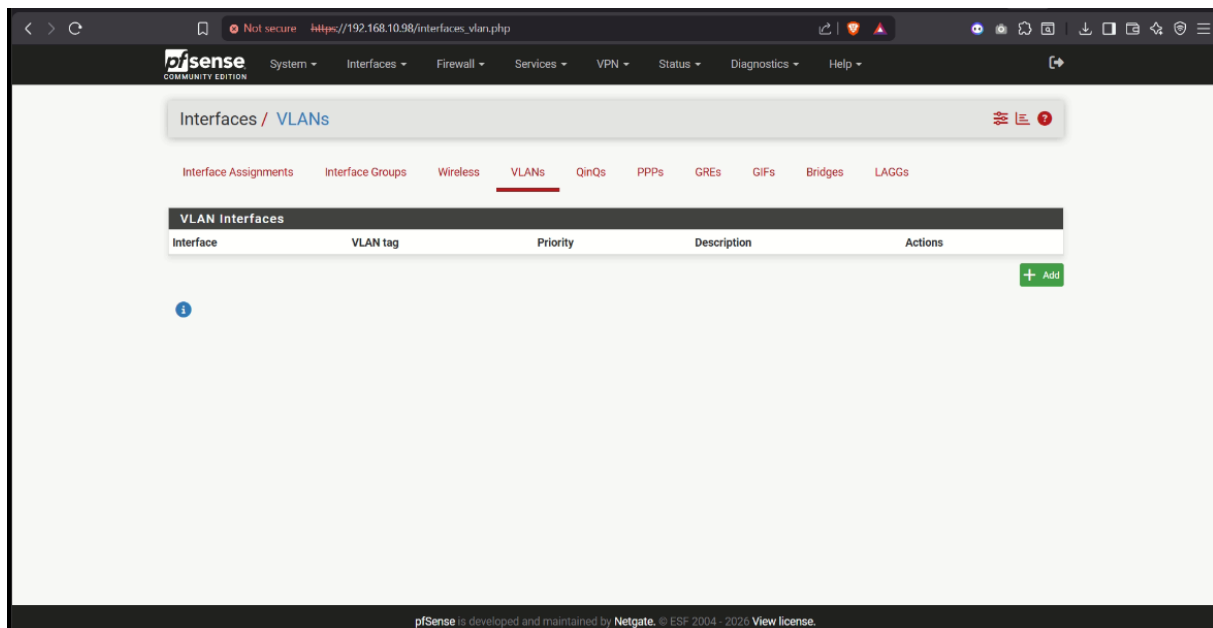
Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☒
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

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13. Accès à la gestion des VLANs : Navigation vers le menu des VLANs pour ajouter des segments de réseau sur l'interface parente em1.



14. Création du VLAN 10 (vlan-Etudiant) : Ajout du réseau virtuel avec le tag 10 sur l'interface LAN (em1).

15. Création du VLAN 20 (CORP_V20) : Ajout du réseau virtuel avec le tag 20 sur l'interface LAN (em1).

Interfaces / VLANs / Edit

VLAN Configuration

Parent Interface em1 (00:0c:29:eb:86:f0) - lan
Only VLAN capable interfaces will be shown.

VLAN Tag 20
802.1Q VLAN tag (between 1 and 4094).

VLAN Priority 0
802.1Q VLAN Priority (between 0 and 7).

Description CORP_V20
A group description may be entered here for administrative reference (not parsed).

Save

16. Création du VLAN 30 (LAN_MGMT) : Ajout du réseau virtuel avec le tag 30 sur l'interface LAN (em1).

Interfaces / VLANs / Edit

VLAN Configuration

Parent Interface em1 (00:0c:29:eb:86:f0) - lan
Only VLAN capable interfaces will be shown.

VLAN Tag 30
802.1Q VLAN tag (between 1 and 4094).

VLAN Priority 0
802.1Q VLAN Priority (between 0 and 7).

Description LAN_MGMT
A group description may be entered here for administrative reference (not parsed).

Save

17. Assignment des interfaces VLAN (OPT) : Après la création des VLANs, nous avons assigné chaque VLAN à une interface OPT (OPT1, OPT2, OPT3) pour permettre leur configuration individuelle.

Not secure https://192.168.10.98/interfaces_vlan.php

System Interfaces Firewall Services VPN Status Diagnostics Help

Interfaces / VLANs

Interface Assignments Interface Groups Wireless **VLANs** QinQs PPPs GREs GIFs Bridges LAGGs

Interface	VLAN tag	Priority	Description	Actions
em1 (lan)	10		vlan-Etudiant	 
em1 (lan)	20		CORP_V20	 
em1 (lan)	30		LAN_MGMT	 

+ Add

Interface has been added.

Interface Assignments **Interface Groups** Wireless VLANs QinQs PPPs GREs GIFs Bridges LAGGs

Interface	Network port
WAN	em0 (00:0c:29:eb:86:e6)
LAN	em1 (00:0c:29:eb:86:f0)
OPT1	VLAN 10 on em1 - lan (vlan-Etudiant)
OPT2	VLAN 20 on em1 - lan (CORP_V20)
OPT3	VLAN 30 on em1 - lan (LAN_MGMT)

Save

Interfaces that are configured as members of a lagg(4) interface will not be shown.
Wireless interfaces must be created on the Wireless tab before they can be assigned.

18. Configuration de l'interface "VLAN_Etudiant" (OPT1) : Activation de l'interface OPT1, avec une adresse IP statique configurée sur 192.168.11.1/24.

Not secure https://192.168.10.98/interfaces.php?id=opt1

System Interfaces Firewall Services VPN Status Diagnostics Help

Interfaces / OPT1 (em1.10)

General Configuration

Enable ☒ Enable interface

Description
Enter a description (name) for the interface here.

IPv4 Configuration Type

IPv6 Configuration Type

MAC Address
The MAC address of a VLAN interface must be set on its parent interface.

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address /

IPv4 Upstream gateway [+ Add a new gateway](#)
If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a **WAN type interface**.
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

19. Configuration de l'interface "CORP_V20" (OPT2) : Activation de l'interface OPT2 avec une adresse IP statique configurée sur 192.168.20.1/24.

Not secure https://192.168.10.98/interfaces.php?if=opt2

isense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Interfaces / OPT2 (em1.20)

General Configuration

Enable ☒ Enable interface

Description
Enter a description (name) for the interface here.

IPv4 Configuration Type

IPv6 Configuration Type

MAC Address
The MAC address of a VLAN interface must be set on its parent interface

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address /

IPv4 Upstream gateway [+ Add a new gateway](#)
If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be 'none'.
Selecting an upstream gateway causes the firewall to treat this interface as a WAN type interface.
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

20. Vérification de la liste des VLANs : Récapitulatif affichant les trois VLANs (10, 20, 30) correctement associés à l'interface parente em1.

Not secure <https://192.168.10.98/interfaces.php?if=opt3>

pfsense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Interfaces / OPT3 (em1.30)

General Configuration

Enable ☒ Enable interface

Description
Enter a description (name) for the interface here.

IPv4 Configuration Type

IPv6 Configuration Type

MAC Address
The MAC address of a VLAN interface must be set on its parent interface

MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IP header size) and minus 60 for IPv6 (TCP/IP header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

minus 60 for IPv6 (TCP/IP header size) will be in effect.

Speed and Duplex
Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address /

IPv4 Upstream gateway [+ Add a new gateway](#)
If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be 'none'.
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐

21. Configuration du serveur DHCP pour "VLAN_Etudiant" : Activation du service DHCP sur l'interface VLAN_ETUDIANT avec une plage d'adresses IP allant de 192.168.11.10 à 192.168.11.100.

Not secure https://192.168.10.98/services_dhcp.php?if=opt1

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Services / DHCP Server / VLAN_ETUDIANT

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN **VLAN_ETUDIANT** CORP_V20 LAN_MGMT

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on VLAN_ETUDIANT interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

LAN **VLAN_ETUDIANT** CORP_V20 LAN_MGMT

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on VLAN_ETUDIANT interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	192.168.11.0/24
Subnet Range	192.168.11.1 - 192.168.11.254
Address Pool Range	192.168.11.10 192.168.11.100 From To The specified range for this pool must not be within the range configured on any other address pool for this interface.

22. Configuration du serveur DHCP pour "CORP_V20" : Activation du service DHCP sur l'interface CORP_V20 avec une plage d'adresses IP allant de 192.168.20.10 à 192.168.20.100.

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

LAN VLAN_ETUDIANT **CORP_V20** LAN_MGMT

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on CORP_V20 interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	192.168.20.0/24
Subnet Range	192.168.20.1 - 192.168.20.254
Address Pool Range	192.168.20.10 From 192.168.20.100 To

23. Configuration du serveur DHCP pour "LAN_MGMT" : Activation du service DHCP sur l'interface LAN_MGMT avec une plage d'adresses IP allant de 192.168.30.10 à 192.168.30.100.

LAN VLAN_ETUDIANT CORP_V20 **LAN_MGMT**

General Settings

DHCP Backend	ISC DHCP
Enable	☒ Enable DHCP server on LAN_MGMT interface
BOOTP	☐ Ignore BOOTP queries
Deny Unknown Clients	Allow all clients When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed in a static mapping on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed in static mappings on this interface will get an IP address within this scope/range.
Ignore Denied Clients	☐ Ignore denied clients rather than reject This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore Client Identifiers	☐ Do not record a unique identifier (UID) in client lease data if present in the client DHCP request This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Primary Address Pool

Subnet	192.168.30.0/24
Subnet Range	192.168.30.1 - 192.168.30.254
Address Pool Range	192.168.30.10 From 192.168.30.100 To

The specified scope for this pool must not be within the scope assigned to any other address pool for this interface.

24. Règles de pare-feu (Firewall Rules) :

- Création d'une règle "Pass" (Autoriser) pour le protocole "Any" sur l'interface LAN_MGMT.

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / Edit

Edit Firewall Rule

Action Pass
 Choose what to do with packets that match the criteria specified below.
 Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule
 Set this option to disable this rule without removing it from the list.

Interface LAN_MGMT
 Choose the interface from which packets must come to match this rule.

Address Family IPv4
 Select the Internet Protocol version this rule applies to.

Protocol Any
 Choose which IP protocol this rule should match.

Source

Source ☐ Invert match Any Source Address /

Destination

Destination ☐ Invert match Any Destination Address /

Extra Options

- Création d'une règle "Pass" (Autoriser) pour le protocole "Any" sur l'interface VLAN_ETUDIANT.

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / Edit

Edit Firewall Rule

Action Pass
 Choose what to do with packets that match the criteria specified below.
 Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule
 Set this option to disable this rule without removing it from the list.

Interface VLAN_ETUDIANT
 Choose the interface from which packets must come to match this rule.

Address Family IPv4
 Select the Internet Protocol version this rule applies to.

Protocol Any
 Choose which IP protocol this rule should match.

Source

Source ☐ Invert match Any Source Address /

Destination

Destination ☐ Invert match Any Destination Address /

Extra Options

- Création d'une règle "Pass" (Autoriser) pour le protocole "Any" sur l'interface CORP_V20.

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / Edit

Edit Firewall Rule

Action Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule
Set this option to disable this rule without removing it from the list.

Interface CORP_V20
Choose the interface from which packets must come to match this rule.

Address Family IPv4
Select the Internet Protocol version this rule applies to.

Protocol Any
Choose which IP protocol this rule should match.

Source

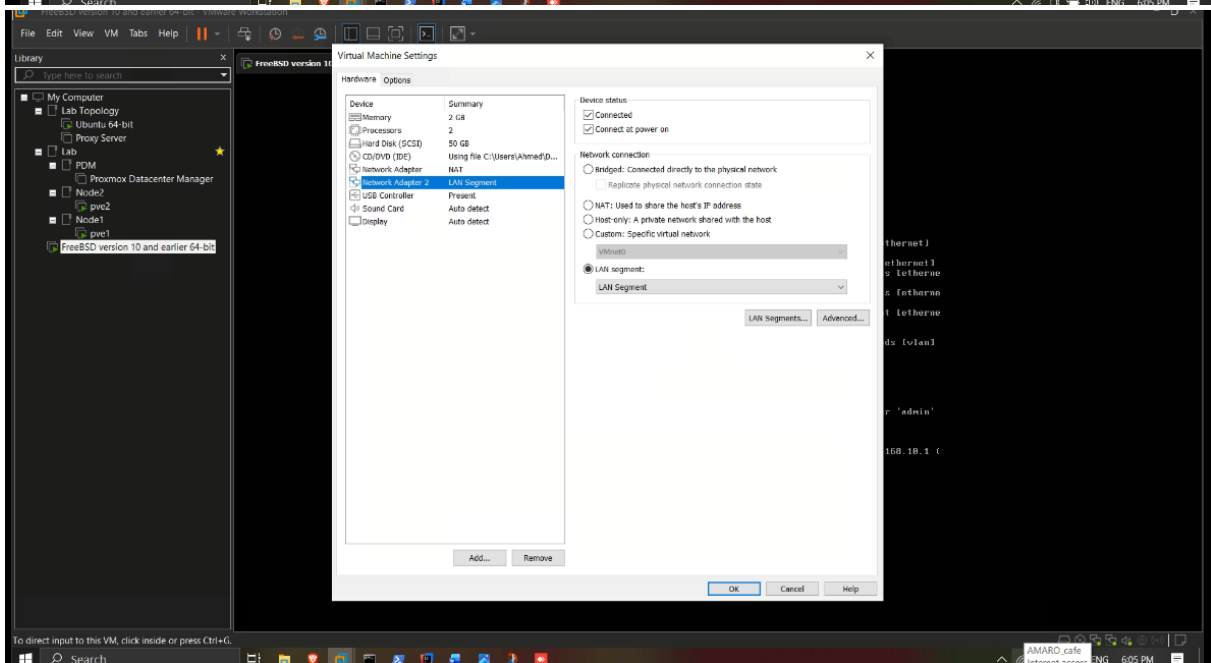
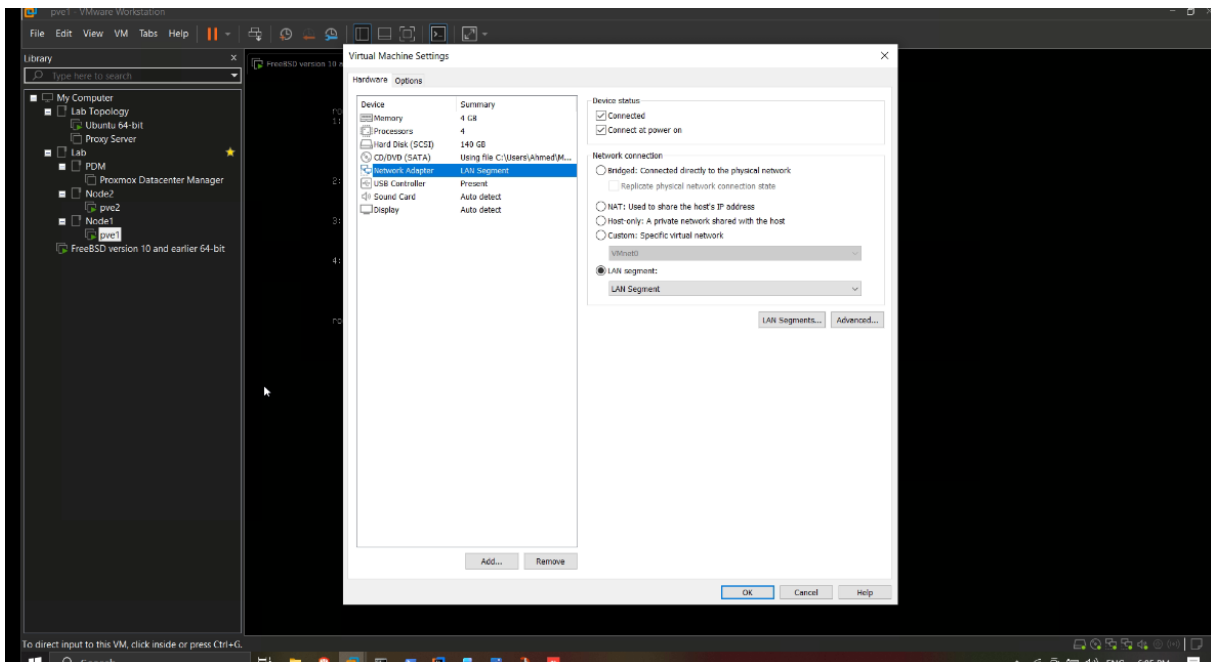
Source ☐ Invert match Any Source Address /

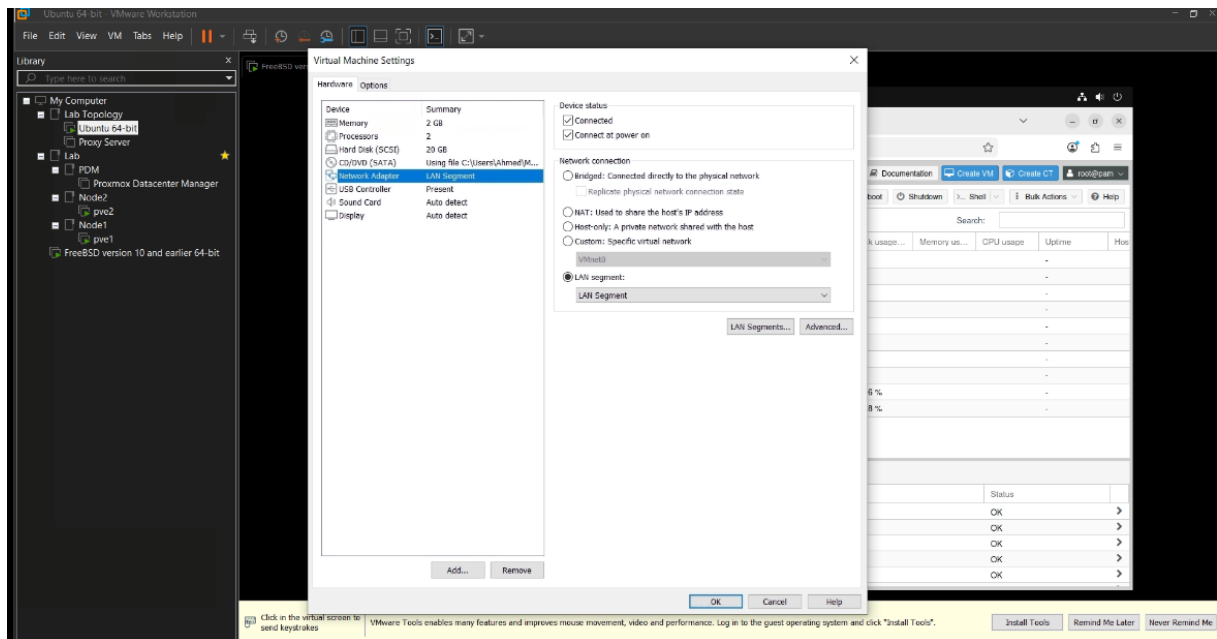
Destination

Destination ☐ Invert match Any Destination Address /

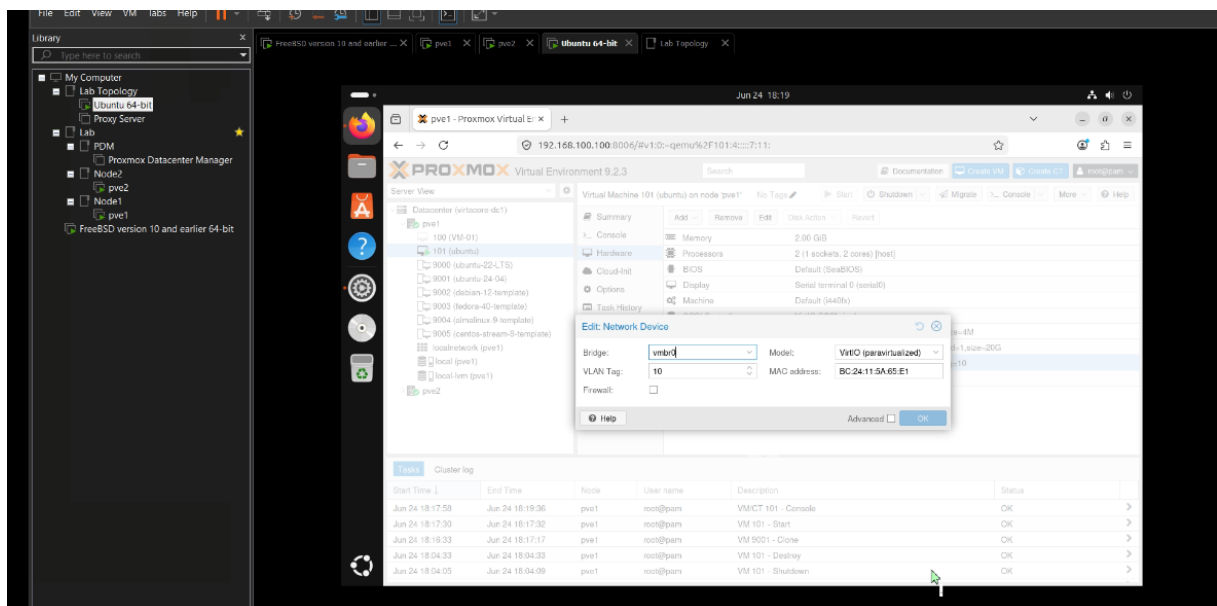
Enter Options

- **25. Configuration du réseau sur Proxmox :** Dans l'interface de gestion Proxmox, nous avons configuré les paramètres réseau de la machine virtuelle Ubuntu (VM 101).





- **26. Affectation du VLAN 10 à la VM :** Nous avons assigné le "VLAN Tag: 10" à la carte réseau virtuelle de la machine Ubuntu sur le bridge vmbr0.



- **27. Test de connectivité et DHCP :** Après le démarrage de la machine Ubuntu, nous avons vérifié l'attribution automatique de l'adresse IP via la commande `ip a`.
- **28. Validation de l'adresse IP :** La machine Ubuntu a reçu avec succès une adresse IP dans la plage 192.168.11.0/24 (attribuée par le serveur DHCP du pfSense), confirmant que le routage VLAN est opérationnel.

